

# Geonu Lee (이건우)

AI Research Engineer

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## Work Experiences

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| Jul. 2024 - Present   | <p><b>Full-time – SNUAILAB</b><br/><i>AI Research Engineer</i></p> <ul style="list-style-type: none"><li>• <b>Led the end-to-end development and deployment</b> of an AI-based smartphone cover glass micro-defect inspection system, achieving a False Negative Rate (FNR) below 2%.</li><li>• <b>Managed technical alignment and crisis resolution</b> with a manufacturing client, establishing technical trust that successfully secured strategic corporate investment.</li><li>• <b>Trained deep learning models</b> for a battery pole tracking system and integrated real-time pipelines into a C++ SDK, stabilizing edge deployment for a specialized inspection equipment manufacturer.</li><li>• <b>First-authored</b> “Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection”, <i>Neurocomputing</i>.</li></ul> |
| Oct. 2023 - Apr. 2024 | <p><b>Full-time – ALCHERA</b><br/><i>AI Research Engineer, Anomaly Analysis Team</i></p> <ul style="list-style-type: none"><li>• <b>Developed and deployed DETR-based object detection models</b> for FireScout, a real-time vision-based wildfire detection solution, achieving a 20%p performance improvement over the legacy baseline.</li><li>• <b>Reconfigured the evaluation pipeline</b> from image-level to instance-level metrics, resolving legacy evaluation limitations to establish a precise and robust performance measurement system.</li></ul>                                                                                                                                                                                                                                                                                             |
| Mar. 2023 - Sep. 2023 | <p><b>Intern – Naver Cloud</b><br/><i>AI Research Engineer, Image Vision Team</i></p> <ul style="list-style-type: none"><li>• <b>Contributed to the development of a face anti-spoofing service framework</b> for eKYC services, constructing a robust system capable of detecting spoofing attacks across diverse environmental conditions.</li><li>• <b>First-authored a paper published at WACV 2025</b> on object anti-spoofing, improving HTER by 7.5% and EER by 6.6% to significantly enhance domain generalization performance.</li></ul>                                                                                                                                                                                                                                                                                                           |
| Jan. 2020 - Feb. 2020 | <p><b>Intern – Electronics and Telecommunications Research Institute (ETRI)</b><br/><i>AI Research Engineer</i></p> <ul style="list-style-type: none"><li>• Developed a human action recognition system that models interactions between humans and surrounding objects in video data.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Research Interests

Deep Learning, Pose Estimation, Multi-Modal, Vision-Language-Model, Vision-Language-Action, Anomaly Detection, Domain Generalization, Face Anti-Spoofing, Human Action Recognition, Human-Object Interaction Detection

## Publications

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- Jun. 2026 | **Geonu Lee**, Yujeong Oh, Geonhui Jang, Soyoung Lee, Jeonghyo Song, Sungmin Cha, YoungJoon Yoo, "Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection," in *Neurocomputing* (Minor Revision)
- Feb. 2025 | **Geonu Lee**, Yonghyun Jeong, Haneol Jang, YoungJoon Yoo, "Domain-Generalized Object Anti-Spoofing: Bridging Gaps and Patch Selection for Robust Detection across Domains," in *Winter Conference on Applications of Computer Vision (WACV)*
- Sep. 2022 | **Geonu Lee**, Kimin Yun, Jungchan Cho, "Occluded Pedestrian-Attribute Recognition for Video Sensors Using Group Sparsity," in *Sensors*, vol. 22, no. 17, pp. 6626
- Aug. 2022 | **Geonu Lee** and Jungchan Cho, "STDP-Net: Improved Pedestrian Attribute Recognition Using Swin Transformer and Semantic Self-Attention," in *IEEE ACCESS*, vol. 10, no.1, pp. 82656 - 82667
- Feb. 2021 | **Geonu Lee**, Kimin Yun, and Jungchan Cho, "Improved Human-Object Interaction Detection through On-the-Fly Stacked Generalization," in *IEEE ACCESS*, vol. 9, no. 1, pp. 34251-34263
- Feb. 2020 | Bhishan Bhandari, **Geonu Lee**, and Jungchan Cho, "Body-Part-Aware and Multitask-Aware Single-Image-Based Action Recognition," in *Applied Science*, vol. 10, no. 4, pp. 1531-1548

## Research Projects

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- 2022 | **A Study on the Complex Human Attributes for Situation Understanding**  
*Master's Research Project – Gachon University*
- Conducted research on recognizing interactive human attributes in video sequences under complex contexts.
  - Utilized Swin Transformer and Transformer decoder architectures for semantic reasoning.
  - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2021 | **A Study on the Understanding of Pedestrians**  
*Master's Research Project – Gachon University*
- Developed a pedestrian attribute recognition method robust to occlusion.
  - Applied group sparsity regularization to handle various levels of visual obstruction.
  - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2020 | **A Study on the Understanding of Human-Object-Interactions**  
*Undergraduate Research Project – Gachon University*
- Designed a novel deep neural architecture based on on-the-fly stacked generalization for HOI detection.
  - Focused on modeling dynamic interactions between humans and surrounding objects.
  - Funded by Electronics and Telecommunications Research Institute (ETRI).
- 2019 | **A Study on the Understanding of Human Situation Based on Deep Learning**  
*Undergraduate Research Project – Gachon University*
- Proposed a multi-task learning framework combining human pose estimation and action recognition.
  - Addressed real-world situational understanding using contextual cues.
  - Funded by Electronics and Telecommunications Research Institute (ETRI).

Education

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|-----------------------|-------------------------------------------------------------------------------|
| Mar. 2021 - Feb. 2023 | <b>Gachon University</b><br><i>M.S. in Software Engineering</i>               |
| Mar. 2016 - Feb. 2021 | <b>Gachon University</b><br><i>B.S. in Department of Computer Engineering</i> |